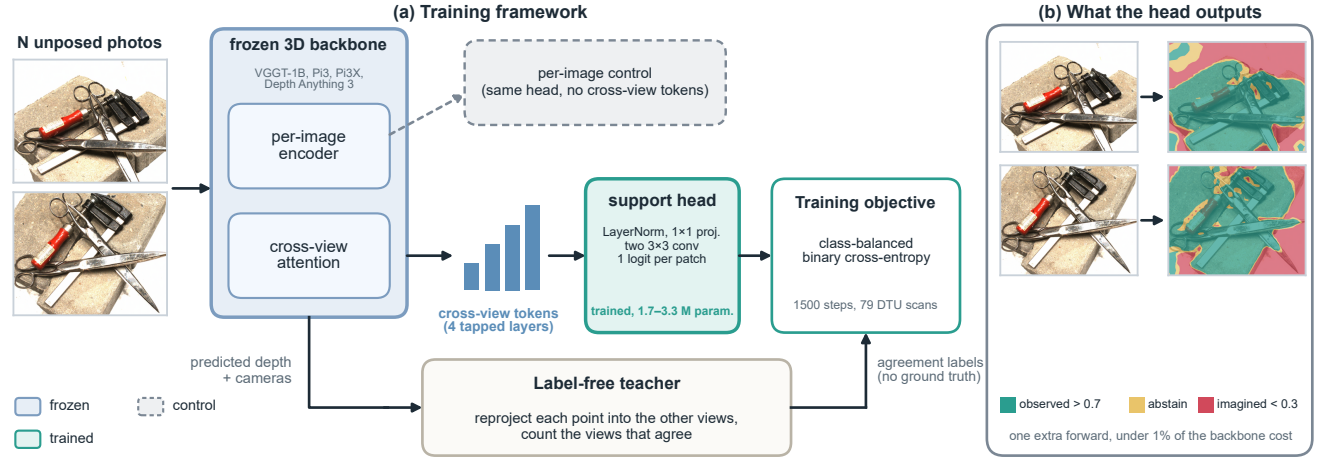

(a) Training framework
N unposed photos
frozen 3D backbone
VGGT-1B, Pi3, Pi3X,
Depth Anything 3
per-image encoder
cross-view attention
per-image control
(same head, no cross-view tokens)
cross-view tokens
(4 tapped layers)
support head
LayerNorm, 1×1 proj.
two 3×3 conv
1 logit per patch
trained, 1.7–3.3 M param.
Training objective
class-balanced
binary cross-entropy
1500 steps, 79 DTU scans
predicted depth
+ cameras
Label-free teacher
reproject each point into the other views,
count the views that agree
agreement labels
(no ground truth)
frozen
control
trained
(b) What the head outputs
observed > 0.7
abstain
imagined < 0.3
one extra forward, under 1% of the backbone cost